



User Manual **SAMmobile RD-150**

Spectroscopic Mobile Radiation
Detection System



Operation Manual

SAMMOBILE 150-D

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CHAPTER 1. INTRODUCTION

FUNCTIONAL DESCRIPTION

The SAMmobile 150-D is a spectroscopic mobile radiation detection system, developed for a vehicle, boat, and aircraft use. The unit includes up to two 2x4x16 (or 4x4x16) inch NaI(Tl) detectors and optional He-3 or tileable, solid-state neutron detectors. The standard configuration places the detector(s) and electronics in a water and shock resistant, reinforced case. Other configurations are available according to specific requirements.

The unit can be deployed in a trunk, on a truck bed, or on a boat deck in a “Load and Go” fashion, while wirelessly reporting events on a separate display.

It can also be used as a temporary portal for pedestrian or vehicle screening.

The default User Interface software for the SAMmobile 150-D is PeakGo-D for Windows-based tablets. As an alternative, the use of SAM III PeakAbout-D app for Smartphones is also possible. Both easy-to-use apps conveniently display isotope names, isotope classifications, color-coded energy peaks, and bar graphs. The rich feature-set in PeakAbout-D and PeakGo-D include GPS tagging, attachment of photo/video images, pop-up isotope help, and immediate ‘Push-to-Reach-Back’ functionality.



Figure 1.1 A typical SAMmobile 150-D application example

IMPORTANT SAFETY INFORMATION

The SAMmobile 150-D has been meticulously engineered to provide safe and dependable high performance for years to come. But, as with any form of electrical equipment, following instructions and taking basic precautions will prevent injury to the operator, or damage to the equipment. Therefore, before using this equipment, the following precautions should be observed:

- ✓ Read and understand this manual thoroughly.
- ✓ Store all provided documents in a safe place for future reference.
- ✓ Be especially sure to read the “SAFETY” chapter of this manual, and thoroughly understand all warning and caution labels.

PRODUCT WARRANTY

Description of the Berkeley Nucleonics Corp. warranty for the SAMmobile 150-D VMRDS (Vehicle Mounted Radiation Detection System):

- ✓ The unit is warranted to be free of defects in material and artistry, including parts, labor, and all unit elements.
- ✓ Unless otherwise indicated, the warranty is good for 12 months after delivery to the original buyer only.

CHAPTER 2. SAFETY

CAUTIONS AND NOTES

The following warnings and instructions are vital for the proper operation of the SAMmobile 150-D unit and should be made readily accessible for reference. Follow these warnings and instructions thoroughly to use the unit correctly and to protect the safety of the operator.



Handling Instructions:

- ✓ Qualified/trained personnel should handle the maintenance service of the unit.



Safety Hazard Warnings:

- ✓ Internal high voltage (high voltage in the detector is up to 1000 Volts) makes it very hazardous to open the unit. Be sure to refer all servicing to qualified/trained personnel.



Disposal Instructions:

- ✓ If/when disposing of the unit: observe local and national regulations for the disposal of units containing lead.
- ✓ If/when disposing of the unit: observe local and national regulations for the disposal of units containing a lithium battery.
- ✓ Do not use household or municipal waste collection services for the disposal of electrical and electronic equipment. EU countries require the use of separate recycling collection services.



WARNING:

- ✓ External equipment intended for connection to signal input, signal output or other connectors, shall comply with relevant EN Standard (e.g., EN 61326-1:2006, EN 301 489-1 V1.9.2 and EN 301 489-17 V2.2.1).
- ✓ If in doubt, contact a qualified technician or your local representative.

SAMmobile 150-D CONTAINS FCC ID: U8D-FB155BC

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

CLASSIFICATION

- ✓ EMC: Comply with EN 61326-1: 2006, EN 301 489-1 V1.9.2 and EN 301 489-17 V2.2.1)
- ✓ FCC Class: Class B Equipment (FCC CFR 47 part 15 subpart B, Section 15.101)
- ✓ The degree of protection against electric shock: Type BF Applied Part
- ✓ The degree of protection against the ingress of water: IP65
- ✓ Equipment not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.

LABELS

The location of the label providing the system power requirements and a serial number of the system is on the side of the carrying case.

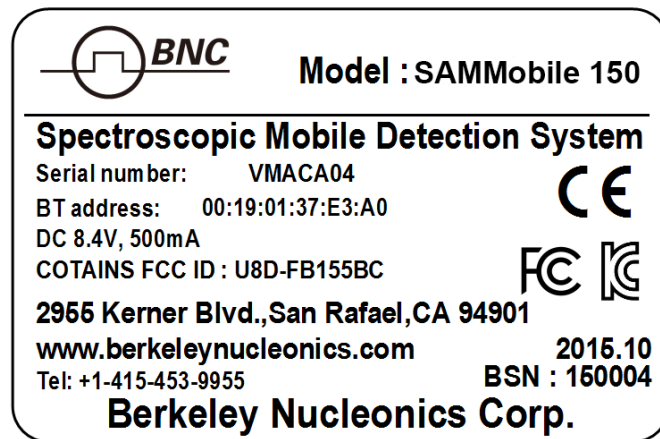


Figure 2.1. Label

CHAPTER 3. SYSTEM DESCRIPTION

SYSTEM COMPONENTS

- ✓ SAMmobile 150-D components, accessories, and carrying case are shown in figure 3.1. Depending on which model is purchased, two possible sizes of carrying case are provided.



Figure 3.1 150-D Components, its accessories and carrying case

- ① SAMmobile 150-D detector (150-D -2G1 or 4G1 model: one detector per case)
- ② SAMmobile 150-D detector (150-D -2G2 or 4G2 model: two detectors per case)
- ③ Tablet PC w/ application SW PeakGo-D loaded
- ④ DC power adaptor for SAMmobile 150-D
- ⑤ 12 V adaptor for vehicle electrical system

POWER AND BATTERY CHARGE GUIDE

- ✓ Figure 3.2 shows the location of the power switch and the DC connector for charging the internal battery. The LED in the power switch turns red when the switch is turned ON.

 The battery charger has a built-in overcharging protection circuit. Once the battery is fully charged, the LED on the battery charger turns green, and auto-charge termination is activated.



Figure 3.2. Power Switch and DC Connector

- ✓ SAMmobile 150-D can also be charged by standard 12 V DC vehicle electric system.

 The SAMmobile 150-D is equipped with a Lithium-Ion battery. The battery has a built-in power protection circuit (PCM) which prevents the battery from both overcharging and over-discharging.

 **Hazard:** Internal high voltage (up to 1,000 Volts)! Do not disassemble the unit.

 Servicing of the unit should be performed by Berkeley Nucleonics Corporation qualified personnel.

SAMMOBILE 150-D SOFTWARE

The SAMmobile 150-D software includes:

1. SAM III PeakGo-D: default application software for the Tablet PC
2. SAM III PeakGo-D Adaptor: adaptor SW between PeakGo-D application SW and SAMmobile 150-D units via Bluetooth connection

The Tablet PC application software is preinstalled in the provided Tablet PC.

PeakGo-D software is used in the configuration and calibration procedures, as well as for the real-time measurement and analysis of spectral data acquired by the SAMmobile 150-D.

The Tablet PC also supports Event Log file management, integration of data from multiple units, and data backup.

For detailed installation instructions, please refer to the appropriate sections in this manual.



Application software for the Tablet PC is compatible with Windows-based Tablet PC.

PREREQUISITES FOR THE OPERATION OF THE SAMMOBILE 150-D



Charge the SAMmobile 150-D battery

1. Turn off the system (recommended for fastest time to full charge)
2. Plug-in the DC adaptor to the DC connector
3. A full charge may take up to 6 hours
4. The LED on the DC power adaptor changes to green when the unit is fully charged.
5. The charging cable is securely locked to the female connector on the SAMmobile 150-D unit. Disengage the spring-loaded connector by firmly holding the male connector and gently twisting $\frac{1}{4}$ turn, then pulling straight back on the connector.



For detailed setup, configuration, background measurement, and calibration procedures, please refer to the appropriate sections in this manual.

CHAPTER 4. TABLET PC APPLICATION SOFTWARE

INTRODUCTION

✓ SAMmobile 150-D Software structure

To connect two detectors simultaneously via Bluetooth to the application SW, a link between hardware devices and application SW is necessary.

Therefore, SAMmobile 150-D software consists of the application SW, PeakGo-D, and the adaptor SW, PeakGo-D Adaptor. The former is the main graphic user interface and connects to the 150-D via PeakGo-D Adaptor. PeakGo-D Adaptor links between the PeakGo-D and 150-D via TCP/IP socket and Bluetooth, respectively. Figure 4.1 illustrates the function and connection of each software.

✓ How to connect SAMmobile 150-D and PeakGo-D

Since the two hardware units have to operate together via the adaptor, the user has to follow the appropriate connection procedure at the beginning and whenever re-connection is needed.

1. Execute PeakGo-D Adaptor SW. (connection to 150-D units).
2. Execute PeakGo-D application SW. (user interface SW).
3. Click  icon on the top menu of PeakGo-D (connection between PeakGo-D and PeakGo-D Adaptor).

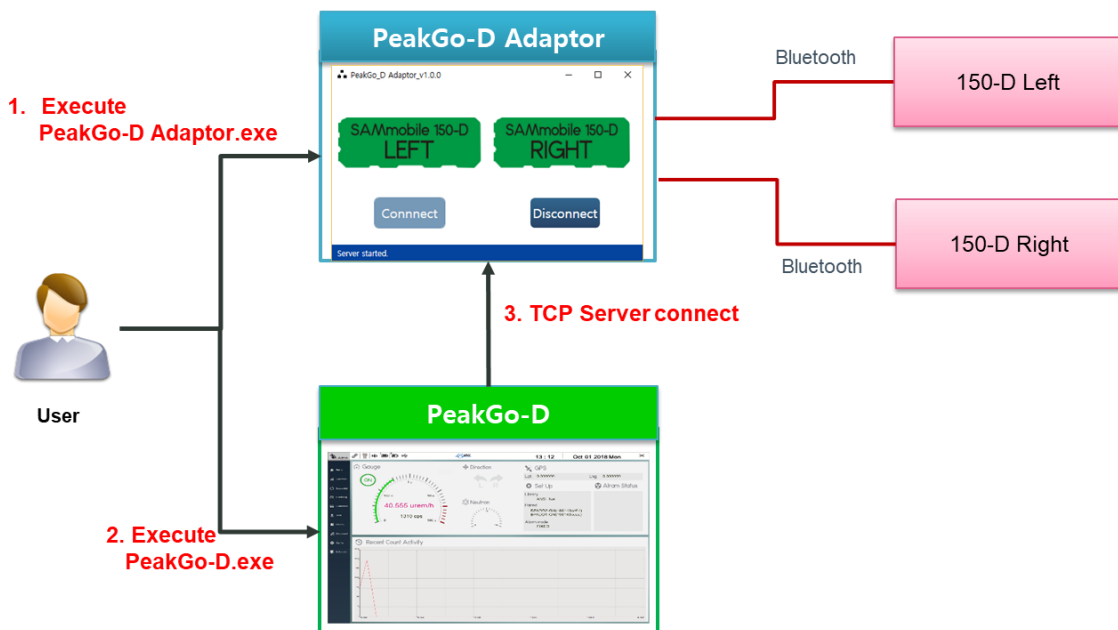


Figure 4.1. PeakGo-D Software Structure

✓ PeakGo-D Application SW GUI

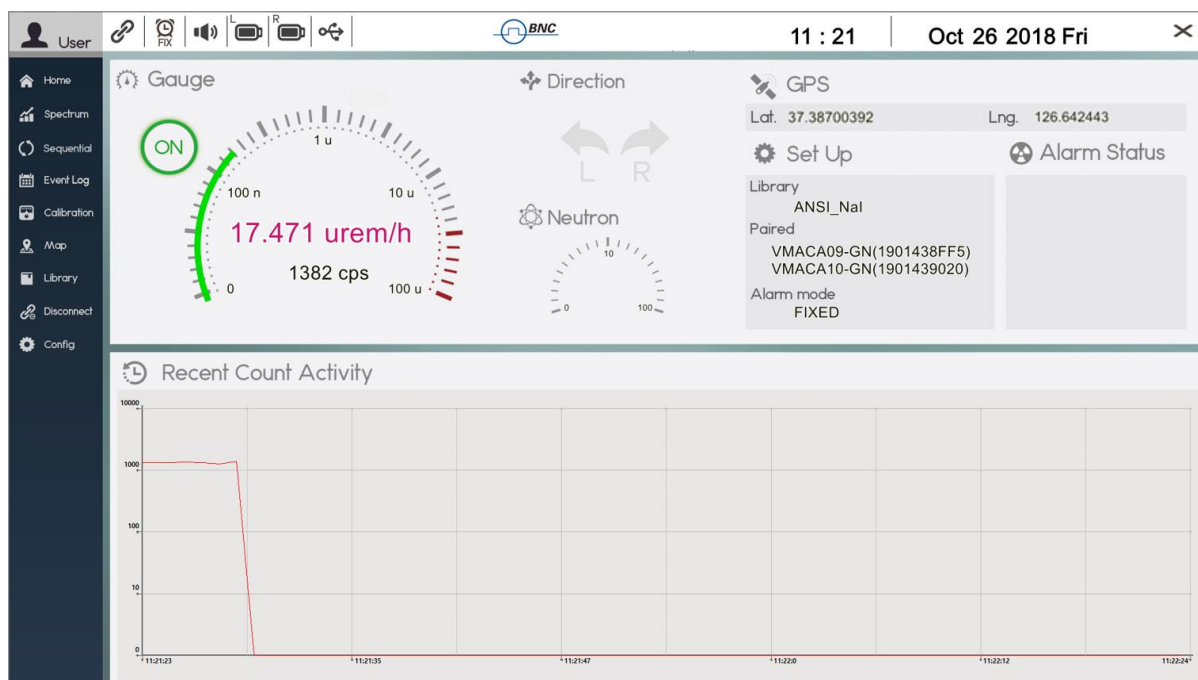


Figure 4.2. Tablet PC Software Main GUI

The Tablet PC communicates with the PeakGo-D Adaptor through the TCP/IP network protocol. Its main function is to provide dose (and count) rate data for gamma-ray energy. Also, it provides the operator with a convenient graphic status display and access to the analytical tools. It also manages the configuration and calibration of the paired SAMmobile 150-D unit(s).

The SW is Windows OS based and compatible with any Windows-based Tablet PCs. The standard unit provided with the SAMmobile 150-D is the Samsung Tablet PC.

The software permits Bluetooth networking only between a Tablet PC and SAMmobile 150-D units; this avoids potential data collision issues with any other Bluetooth devices in the area.

The latest software can be downloadable from <http://bncsam.com/PeakGo-D.html>.

INSTALLATION AND UNINSTALLATION GUIDE

✓ Installation

Download the latest software from <http://bncsam.com/PeakGo-D.html> by following these steps.

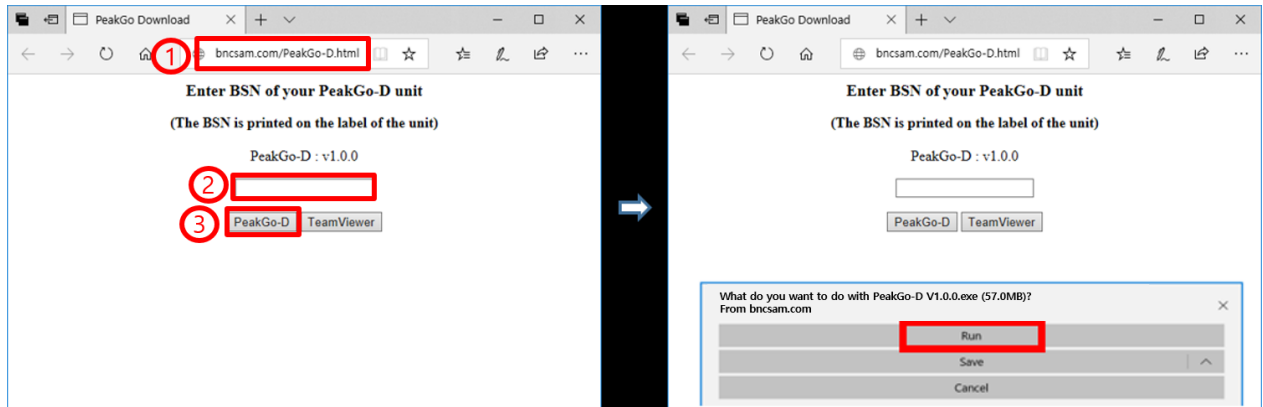


Figure 4.3. PeakGo-D download Example

1. Connect to <http://bncsam.com/PeakGo-D.html>
2. Enter password (the factory saved password is the serial number of the unit. One may find the serial number on the label)
3. Click “Download.”
4. Click “Run” to proceed.
5. Follow the installation steps below.

To install the Tablet PC application software to the Tablet PC, please follow the easy steps below.

1. Download or Copy “PeakGo-D” Setup program from the installation USB drive provided with the unit.
2. Double click “PeakGo-D Setup” file (refer to figure 4.4 below)
3. Follow the easy steps in figure 4.4 to install “PeakGo-D” to the Tablet PC

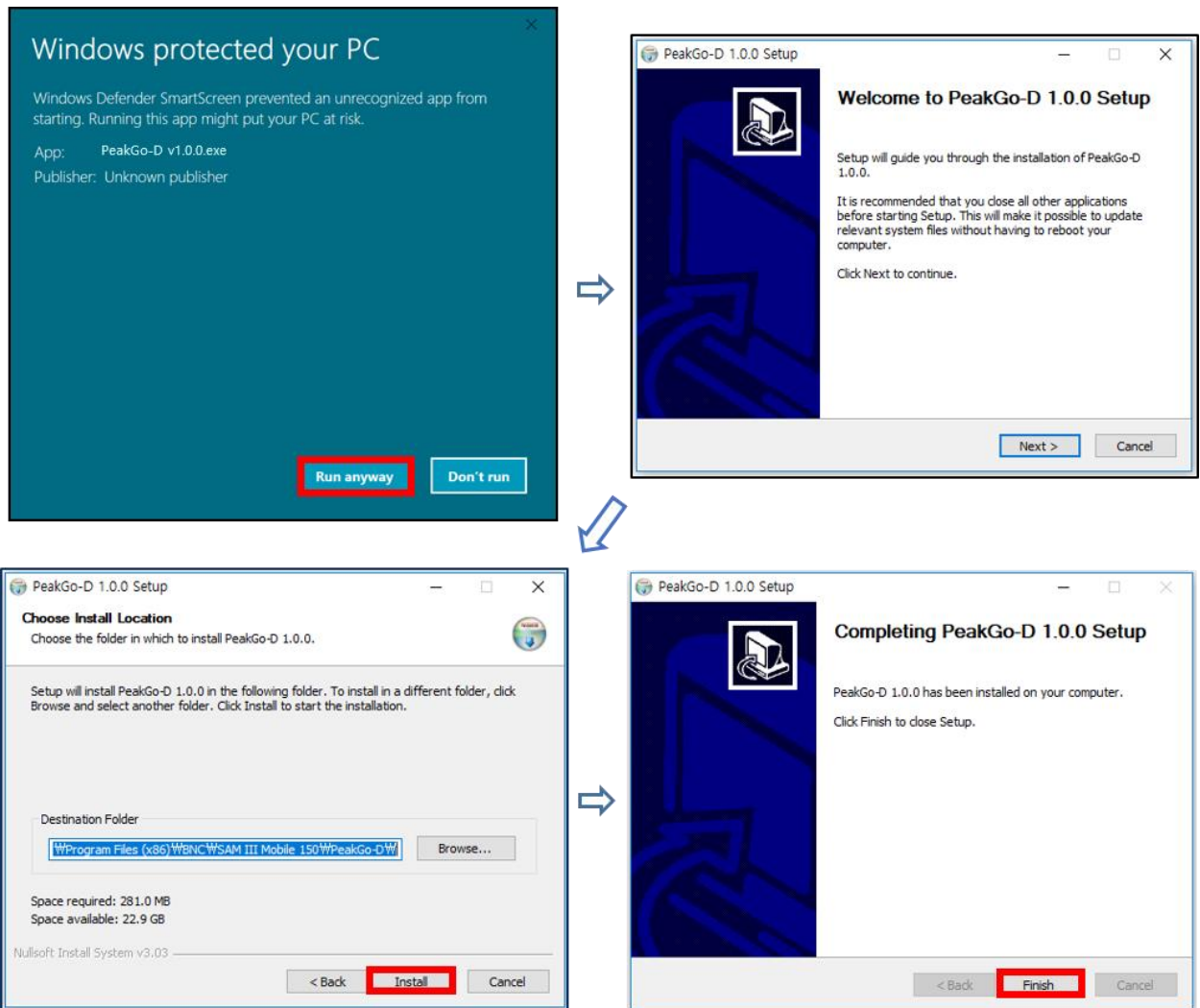


Figure 4.4. PeakGo-D Installation Example

✓ Uninstallation

To uninstall the Tablet PC application software from the Tablet PC, use Windows Control Panel – Programs – Uninstall a program.

1. Select “PeakGo-D”
2. Uninstall “PeakGo-D” (refer to figure 4.5)

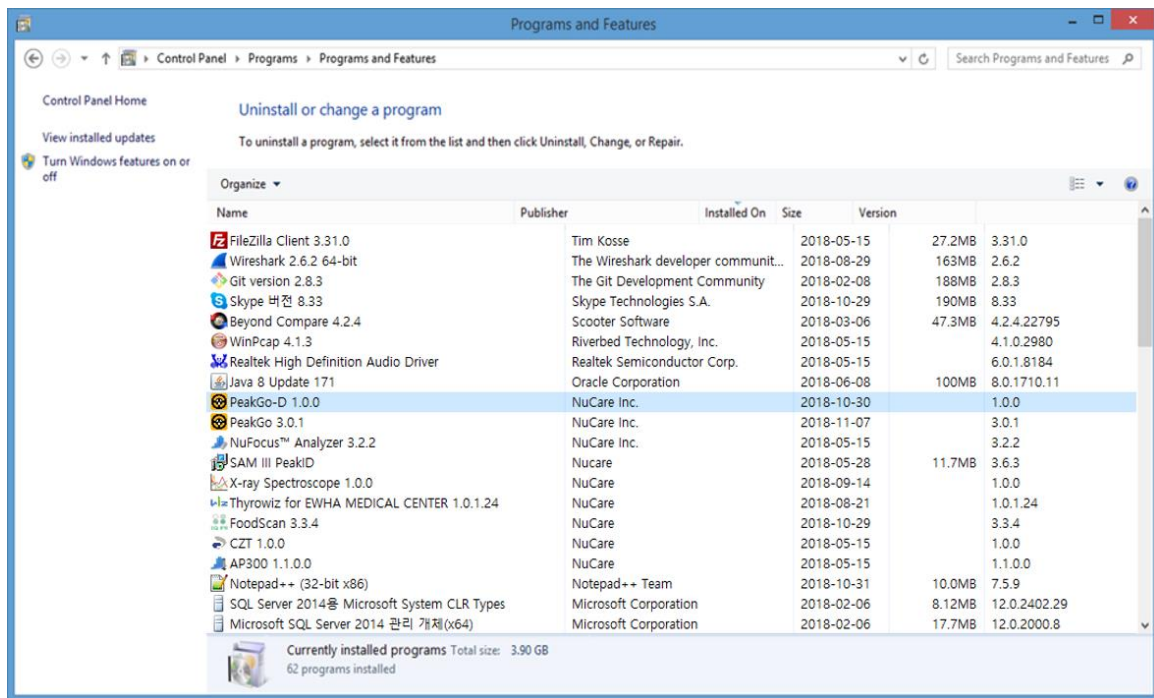


Figure 4.5. PeakGo-D Uninstallation Example.

✓ PeakGo-D Adaptor

PeakGo-D Adaptor provides connect and disconnect functions between SAMmobile 150-D and PeakGo-D application software. Follow the instructions below to manage the adaptor.

- To start PeakGo-D Adaptor

Double-click the executable file,  **PeakGo_Adaptor.exe**, and both SAMmobile 150-D (left and right units) are connected automatically via TCP/IP protocol.

Figure 4.6 shows when the connection is completed. The color of both left and right SAMmobile 150-D icons turn to green.

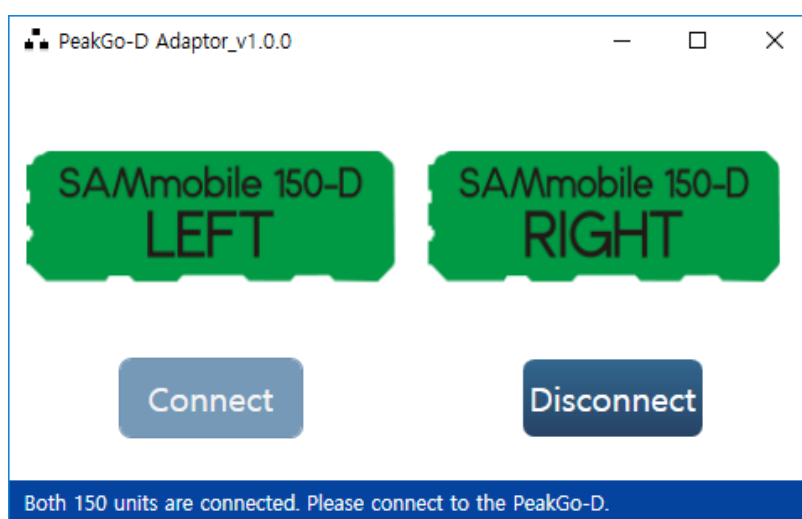


Figure 4.6. PeakGo-D Adaptor Auto Start

 Don't forget to click the  |  |  |  |  network connection icon on the PeakGo application SW. Once the connection is successful, the icon turns to



- Re-connect PeakGo-D Adaptor

If at least one SAMmobile 150-D unit is disconnected, a connection failure message pop-up appears on the PeakGo-D software as shown in figure 4.7.

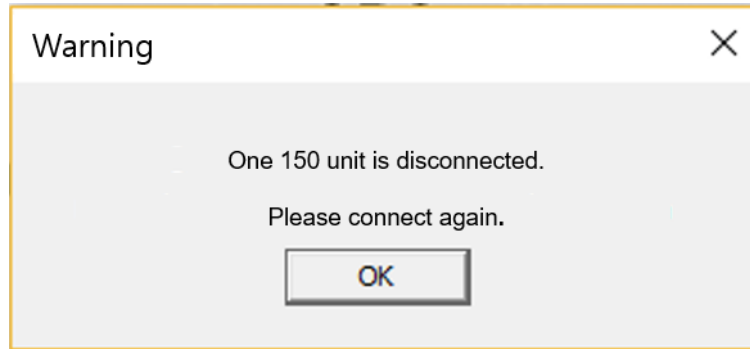


Figure 4.7. Disconnection error message on PeakGo-D

Click the OK button and the adaptor SW window appears as shown in figure 4.8.

1. Click the Connect button.

2. Click the network connection icon  on the PeakGo-D.

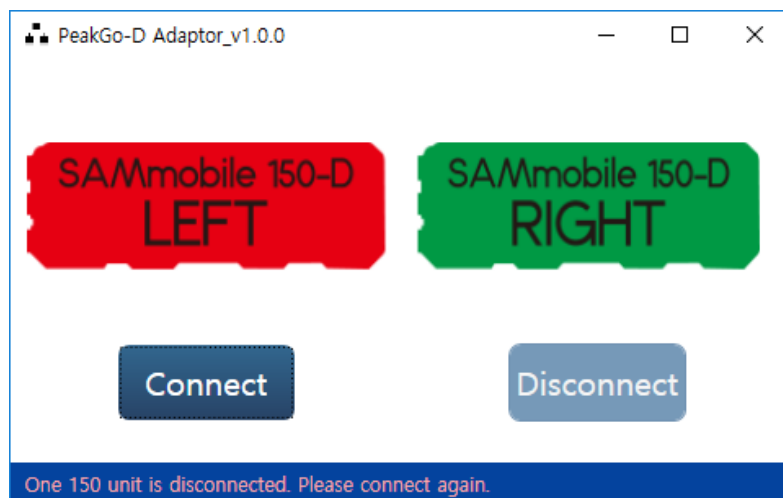


Figure 4.8. PeakGo-D Adaptor re-connection

✓ PeakGo-D main GUI

Measurement status, visual display, spectrum analysis, background measurement, configuration, and calibration are all managed by the Tablet PC and its application software.

Figure 4.9 shows the opening page of the “PeakGo-D” GUI software. The top bar contains Short-Cut Icons and Time and Date. The left side of the primary GUI is assigned for the main Menu. The top half of the Main Screen is assigned for graphical display of dose rate, SAMmobile 150-D monitoring status, GPS data, current configuration data, and alarm status. The bottom half of the Main Screen is a graphical display (histogram) of recent activity in real-time.

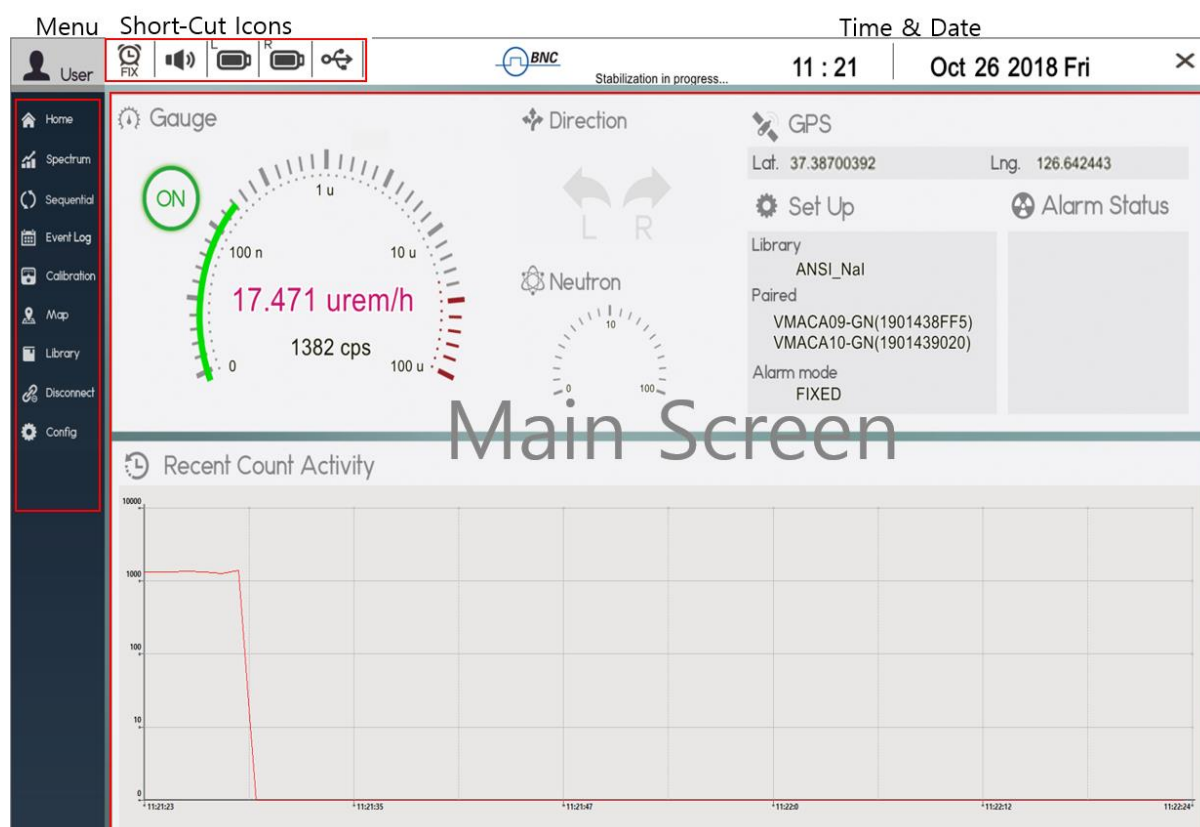


Figure 4.9. PeakGo GUI.

HOME SCREEN

The Home screen and its components are shown in figure 4.10. Details are provided in the following sections.

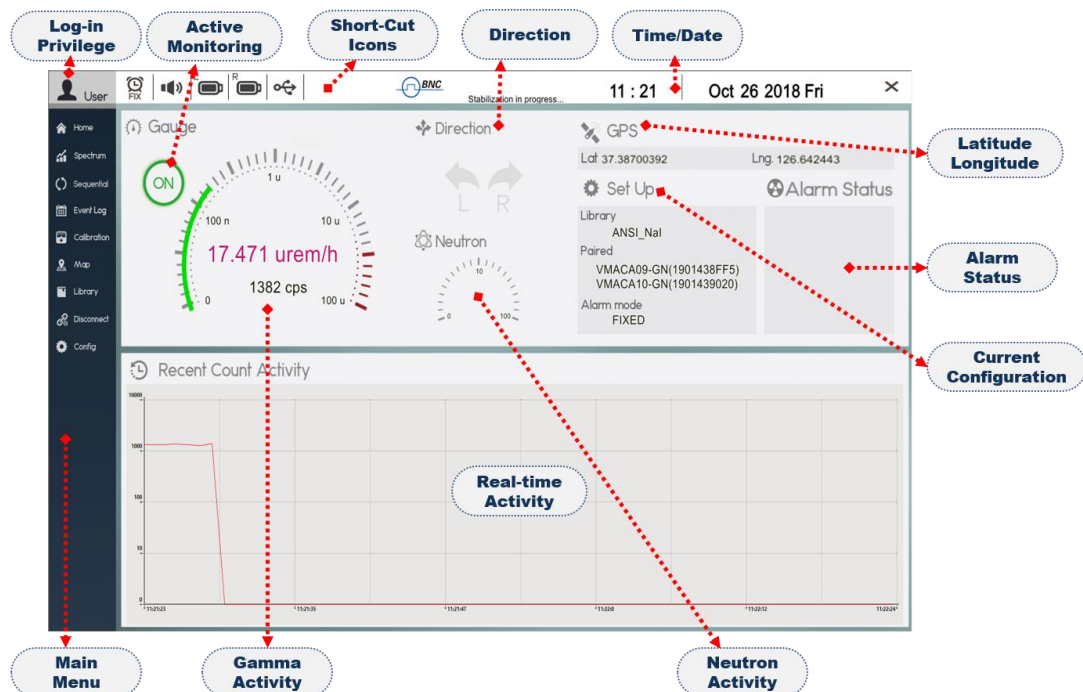


Figure 4.10. Home Screen Description

Main Menu	Allows the operator to choose among corresponding processing screens
Short-Cut Icons	Groups frequently used functions
Log-in Privilege	Displays Log-in status: either Admin or User
Active Monitoring	This displays monitoring status. Allows operator to turn Gauge on/off.
Date/Time	Displays the current date and time information.
Direction	The direction of detected gamma radiation (Left or Right)
GPS	Displays latitude and longitude of the 150-D location
Set Up	Displays current configuration of Library, Paired 150-D, and Alarm mode
Alarm Status	Displays graphical, color-coded icon for alarm status
Neutron	Displays neutron rate in counts per second
Gauge	Displays gamma dose rate and count rate
Recent Count Activity	Displays a histogram to represent gamma activity in real-time that can alert you to a momentary increase in activity faster than the rapidly changing gauge numbers

SPECTRUM SCREEN

Figure 4.11 shows the Spectrum screen analysis features. A spectrum of detected isotopes is displayed on the top half of the screen while qualitative analysis and isotope ID summary are displayed on the bottom half of the screen.

By selecting from the Configuration menu, isotope identification can be initiated automatically or manually. “Manual ID” functions are shown in figure 4.11. Measurement time can be extended/reduced in increments of 30 seconds by pressing the ◀ or ▶ button at the bottom of the screen. ID results are shown as a bar graph with qualitative information and color-coded into 5 different classifications. The Information window shows a list of identified isotopes with corresponding confidence levels.

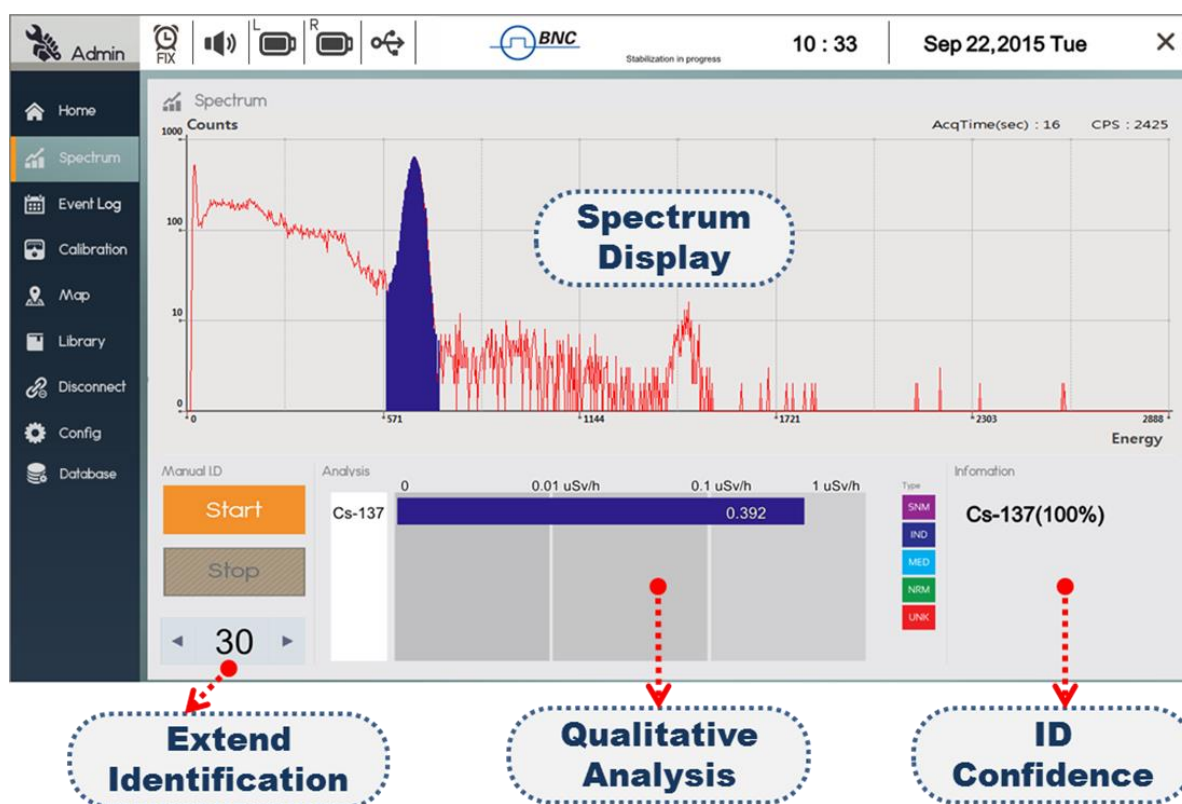


Figure 4.11. Spectrum Screen Description

Spectrum	Spectrum is displayed in the top half of the screen. Live acquisition time and count rate information are displayed at the top of the graph
Manual ID	The bottom left corner has Manual ID Start/Stop button. Measurement time can be extended or reduced by increments of 30 seconds.
Analysis	Further analysis of individually identified isotopes is graphically displayed.
Information	Identified isotopes are summarized with statistical confidence

SEQUENTIAL MODE

The Sequential Mode provides multiple measurements of events sequentially and automatically.

1. Select “Sequential” from the menu and the “Sequential mode” configuration pop-up window appears as shown in figure 4.12.
2. Type in Measurement Time, Pause Time and Number of Measurements (repetition).



You must be logged in as Admin to setup/change Sequential mode parameters.

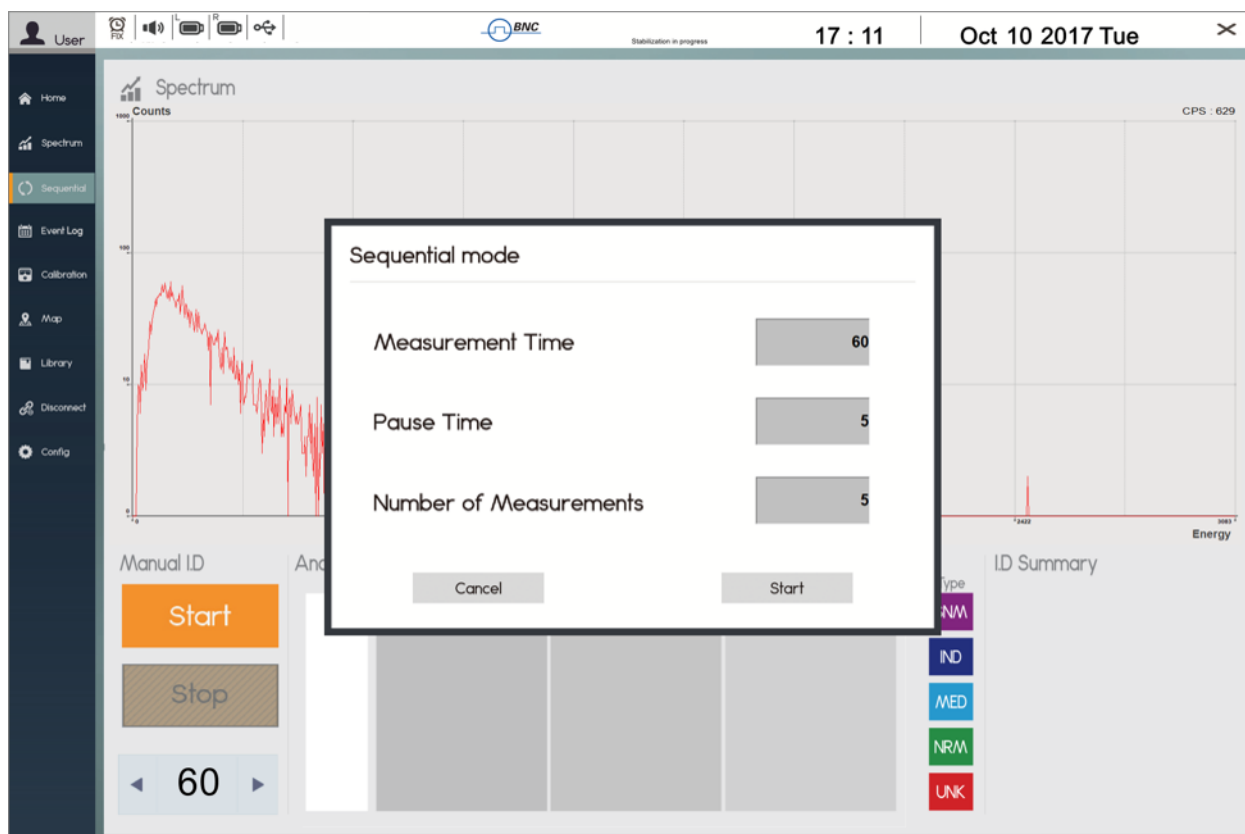


Figure 4.12. Sequential Screen Description

EVENT LOG SCREEN

Figure 4.13 shows the Event Log screen. Events are grouped in the database (DB) by date. To retrieve a saved event, first select from Event by Date. Next, select the event by number or by time stamp from the Event Log list as shown in figure 4.13.

++Each event contains a spectrum, general information, analysis results, photos/videos, and map information. The information is accessed by selecting the corresponding tab on the Event Menu Bar in figure 4.13.

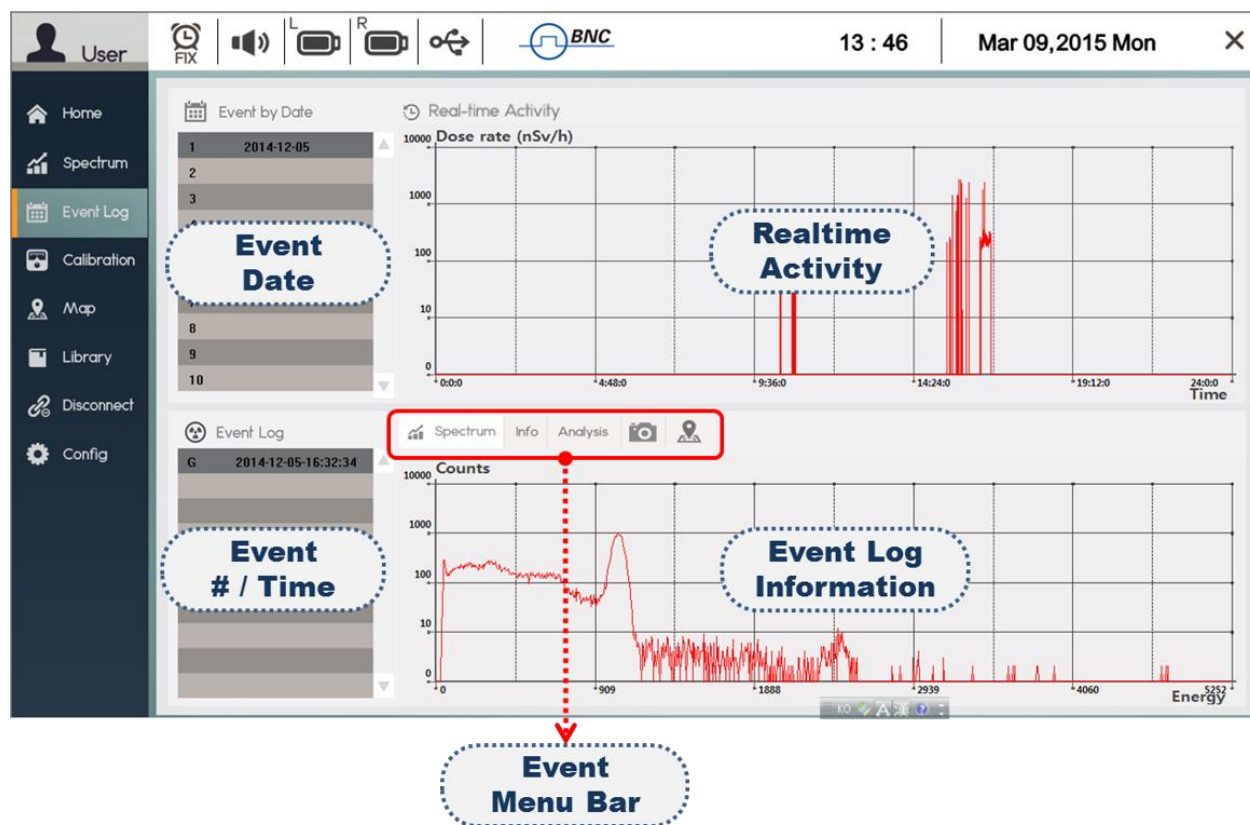


Figure 4.13. Event Log Screen Description

Event by Date	Displays saved events in the database (DB) by date
Event List	Displays saved events on the selected date by event number and/or event time
Realtime Activity	Displays a histogram of real-time activity on the selected date
Event Information	Displays saved information grouped by tabs on the Event Menu Bar

CALIBRATION SCREEN

The Calibration screen consists of a spectrum display window, a calibration/background measurement menu, and a calibration/background procedure window as shown in figure 4.14.

Calibration or background measurement is initiated by clicking the corresponding button located in the left bottom corner as shown in figure 4.14. Measurement parameters are pre-selected and may be adjusted within the Configuration menu.

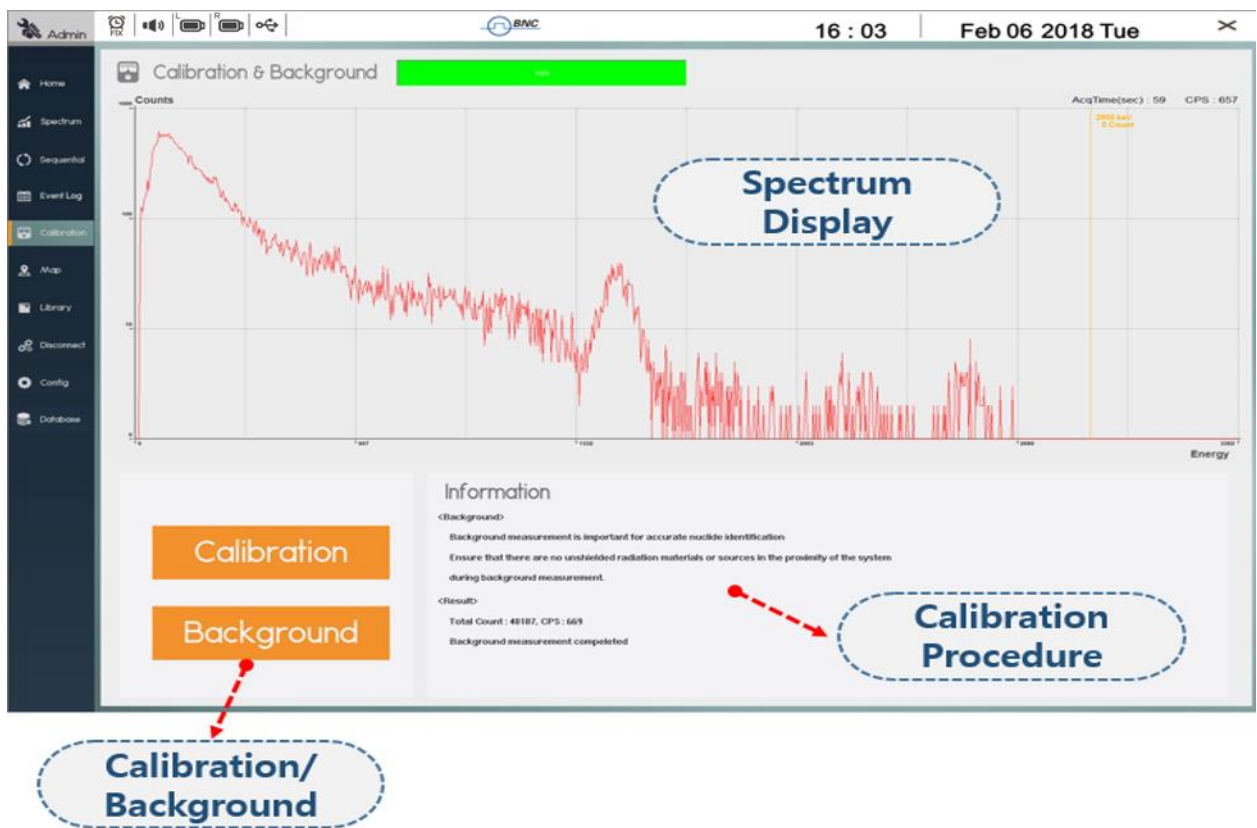


Figure 4.14. Calibration and Background Measurement Screen

Calibration & Background	Displays spectrum of calibration or background during measurement
Calibration/Background	Menu buttons to begin calibration or background measurement
Information	Displays calibration and background measurement procedures and results

MAP SCREEN

If the PC/Tablet is within range of an internet-enabled network, the Map screen displays an area map of the current SAMmobile 150-D location based on GPS information from the Tablet PC. Google Maps provides the area map. The radiation activity detected by the SAMmobile 150-D at the current latitude and longitude is displayed on the map.

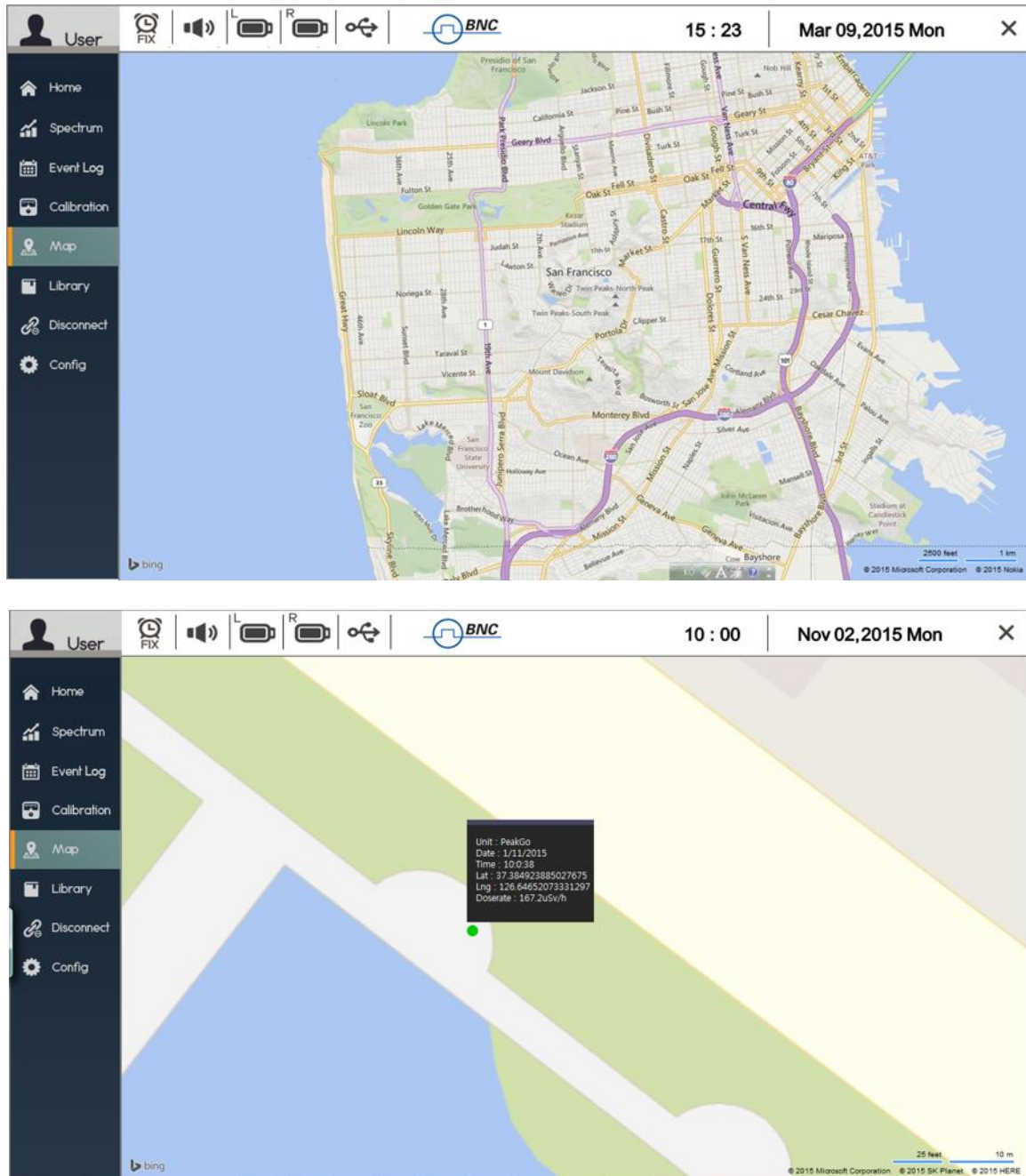


Figure 4.15. Map Screen Description

LIBRARY SCREEN

The Library screen allows the operator to select and read the contents of the library of isotopes. The Library screen consists of three windows, i.e., Library List, Isotope List, and Library Editor as shown in figure 4.16.

The Library List contains the names of standard and/or custom isotope libraries that are used by the SAMmobile 150-D for identification of nuclides. The operator can select/delete a library by using the button below the window. The middle window lists the isotopes that belong to the selected library. In extremely rare cases, the isotopes can be added, deleted, or changed in the Library Editor by using a factory authorized password.

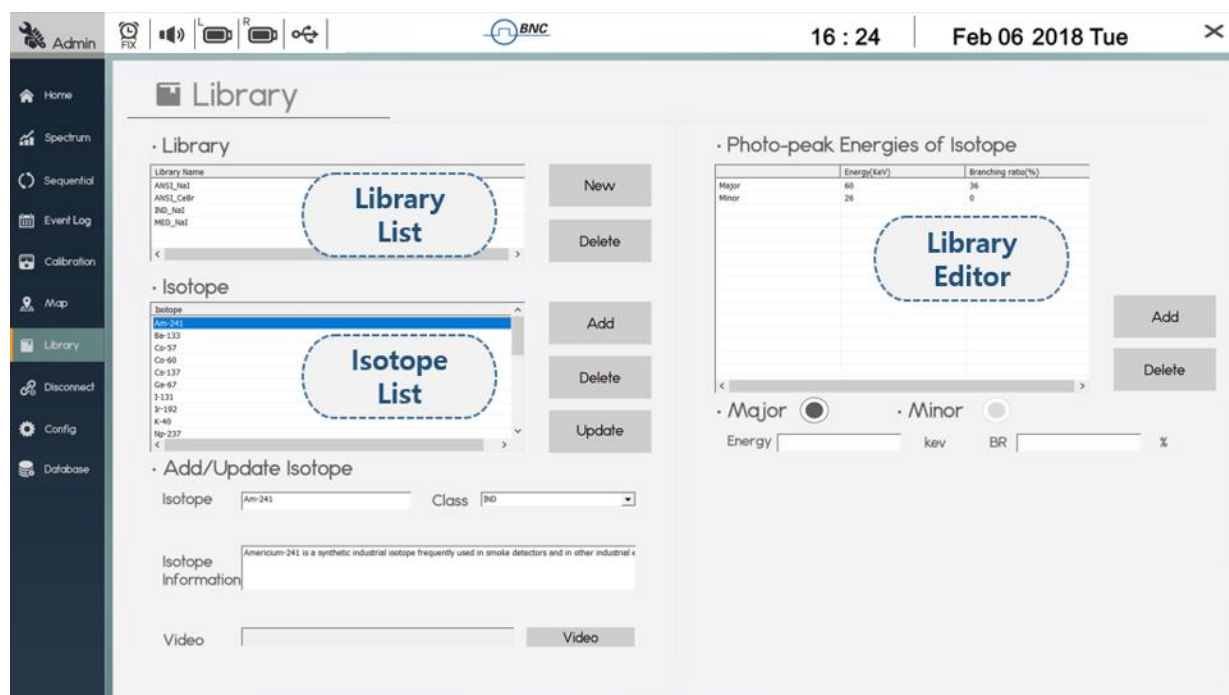


Figure 4.16. Library Screen Description

Library List	Lists available libraries used by the 150-D for isotope identification
Isotope List	Lists isotopes that belong to the selected library
Library Editor	Isotopes can be added/deleted using the edit window (password protected)

CONFIGURATION SCREEN

The Configuration screen (Config) contains a menu of selectable parameters that define User Name/Search Location, Gain Stabilization setup, Background Measurement Time, Calibration Count, Library, Dose Rate Unit, and Alarm setup.

The screenshot shows the Configuration screen of the SAMmobile RD-150 application. The interface has a dark blue sidebar on the left with icons for Home, Spectrum, Sequential, Event Log, Calibration, Map, Library, Disconnect, Config (highlighted), and Database. The main area is titled 'Configuration' and contains several sections with input fields:

- User**: User Name, Search Location
- Gain Stabilization**: Background Limit (CPS), K40 minimum (Count)
- Background**: Measurement Time
- Energy Calibration**: Count (Unit: n x 1000)
- Select Isotope Library**: Library List
- Unit**: Dose Rate Unit
- Alarm**: Set mode (Fixed Alarm Mode, Variable Alarm Mode), Health Safety Threshold, Neutron Threshold (CPS), Alarm Ringtone
- About**: Device Information (More button)
- E-Mail**: Edit button

Figure 4.17. Configuration Screen

 Making changes to the selectable parameters on the Configuration screen is discussed in the Step-by-Step Operation Instructions.

DATABASE SCREEN

Database menu provides a feature of open/backup/extract event logs from storage. The function of each Database submenu is described in the table below Figure 4.18.

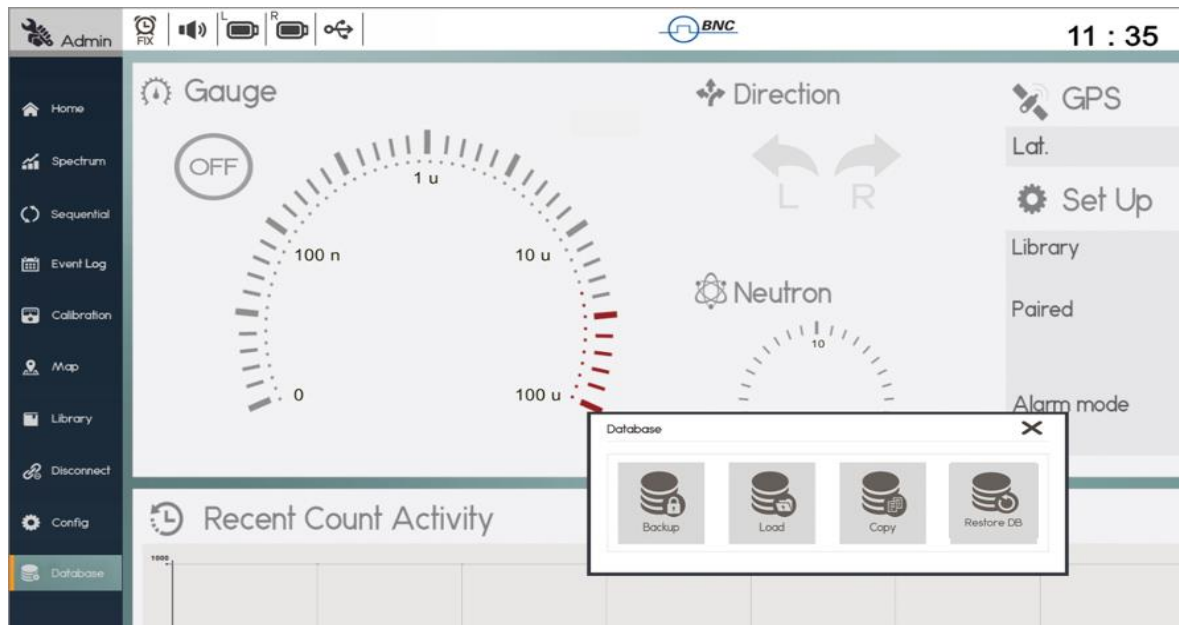


Figure 4.18. Database Screen Description

Backup	Backup the current Event DB to a storage location on the Tablet PC, and delete the current DB from the Event Log screen
Load	Open an Event DB in the Event Log screen from a storage location on the Tablet PC
Copy	Backup the current Event DB to the storage location without deleting from the Event Log screen
Restore DB	Backup an opened Event DB to the storage location and restore the previously current DB to the Event Log screen

SPECTRUM MANIPULATION

✓ Zoom In/Out

The operator can “zoom in” on the spectrum by double-clicking (or double tapping) the area of interest. By double-clicking (or double tapping) on the graph area again, the normal scale is restored.



Figure 4.19. Spectrum “Zoom in” Example

✓ Moveable Reference Line

By clicking or touching the x-axis, the operator can initiate a moveable cursor at the touched position that displays energy and a count value for that energy. The vertical, yellow reference line can be moved to any location on the energy axis by dragging horizontally along the x-axis.

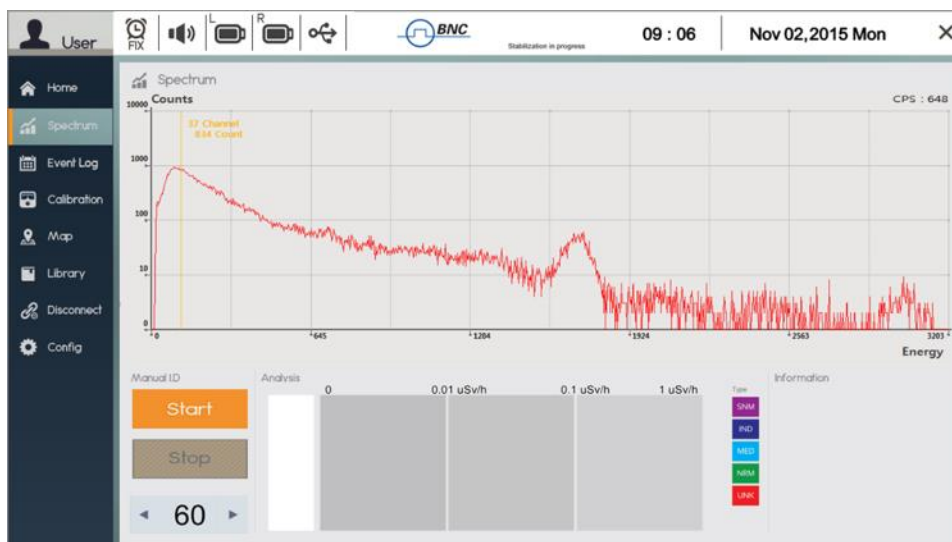


Figure 4.20. Reference Line Example



The count range and maximum count value automatically adjust to accommodate the channel with the highest count.

STEP BY STEP OPERATION INSTRUCTIONS

Step 1. Prepare the SAMmobile 150-D for operation

1. Turn power ON (both –150-D units and the Tablet PC)
2. From the Tablet PC, touch the SAM III PeakGo-D icon.



Step 2. Log In

1. Log in as User or Administrator (password protected).

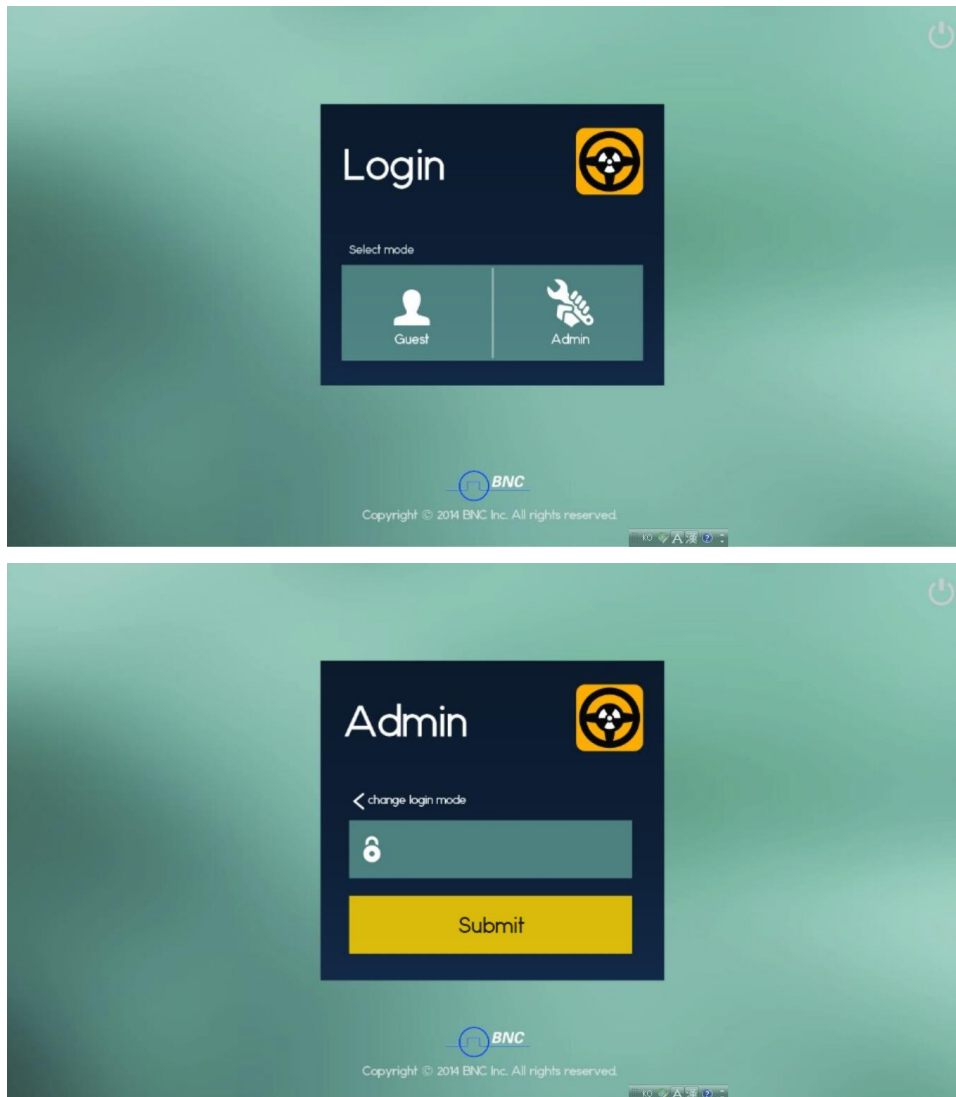


Figure 4.21. Login Screen



A User operator has restricted access to information and setup menus.



The factory default password for Administrator log-in is '1234'. To change the password, go to the Configuration menu - Config/About/Device Information/More/Password edit - as shown in figure 4.22.

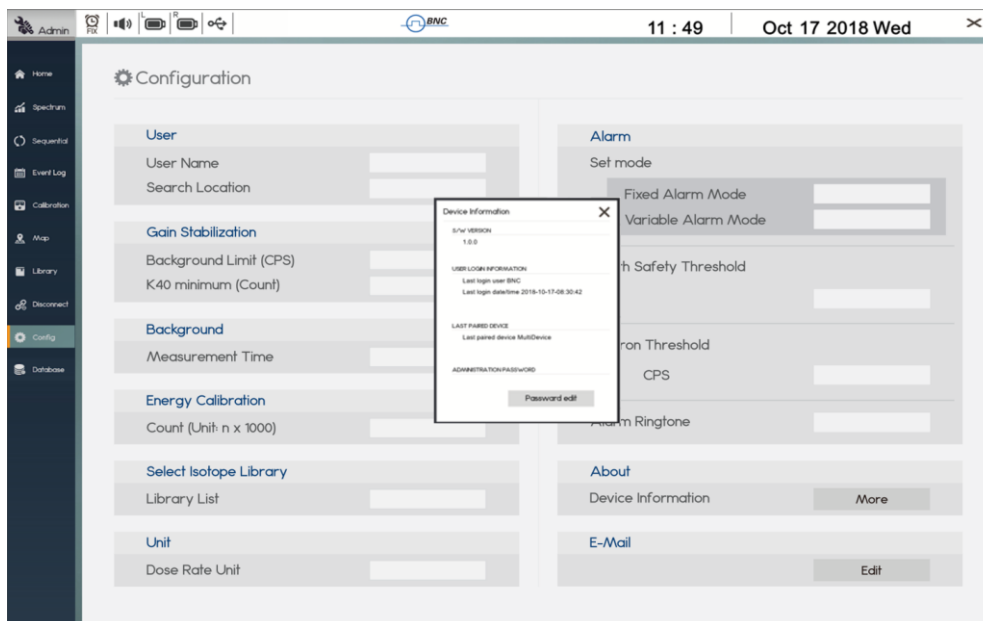


Figure 4.22. Change Password

Step 3. Establish Bluetooth connection between Tablet PC and 150-D

1. Execute PeakGo-D Adaptor.exe(📄).
2. Click 'Connect' button on the PeakGo-D Adaptor
3. Click 'network connection' icon on the short-cut icon menu bar
4. Turn Active Monitoring ON.

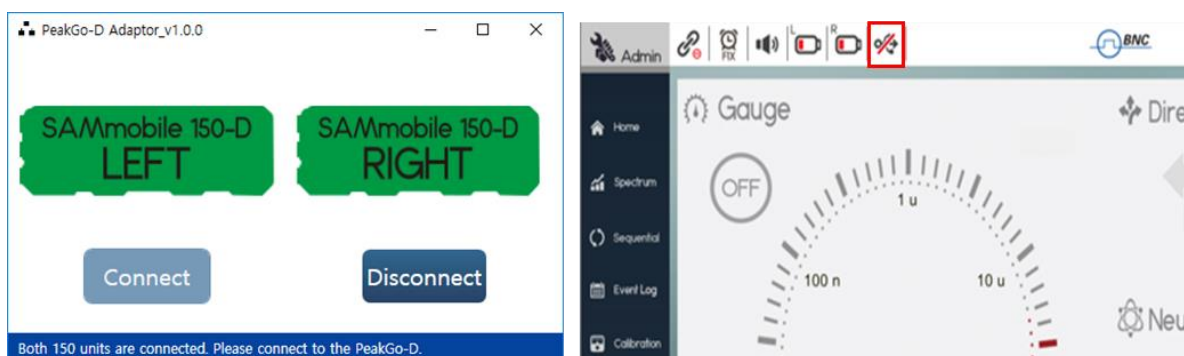


Figure 4.23. Bluetooth Connection Example

 The BT address of the 150-D system is a unique 12-digit code. The 12-digit code and Serial number are found on the label located on the side of the 150-D carrying case (Figure 4.24).

 Although it is a rare case, if you want to connect to a different unit, disconnect the currently paired device by clicking the “Disconnect” button and follow the new Bluetooth connection procedure.

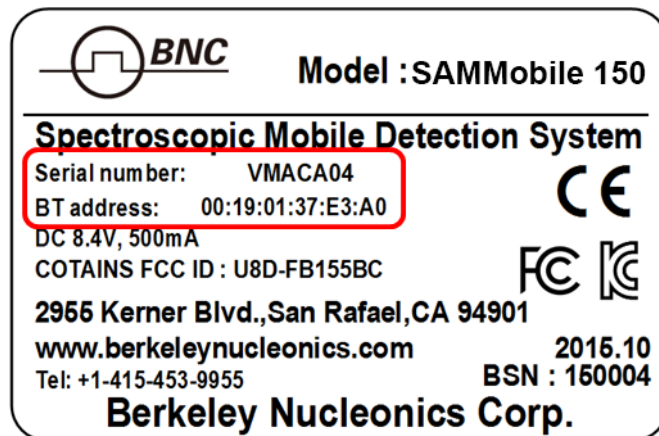


Figure 4.24. BT Address and Serial Number

Step 4. Config: User Name and Search Location



Note: the SAM III PeakGo-D software will not operate if Configuration has not been completed.

1. Click “Config” menu from the main Menu bar
2. Type in “User Name” and “Search Location”

Figure 4.25. Example of User Name and Search Location

Step 5. Config: Alarm Threshold

PeakGo provides two modes for triggering the SAMmobile 150-D alarm. Both are described in the following table.

Fixed Alarm Threshold	This basic method assumes that the background is very stable and exhibits a very small amount of fluctuation. The alarm turns ON when the count rate exceeds the selectable Fixed Alarm threshold value. This method can cause false alarms if the NORM background fluctuates substantially (often the case when the SAMMobile 150-D is moving across a large area while actively monitoring)
Variable Alarm Threshold	This advanced method continuously samples the previous 10 seconds worth of background fluctuation and continuously calculates the standard deviation (σ or sigma) of the fluctuation. The alarm turns ON when the sigma of the fluctuating background exceeds the sigma of the Variable Alarm threshold value. This method is recommended when the system is moving across a large area while actively monitoring (and therefore, subject to fluctuating background).

1. Select Config / Alarm /Set mode (see Figure 4.26)
2. Select either “Fixed Alarm Mode” or “Variable Alarm Mode.”
3. For “Fixed Alarm Mode,” select a count rate value from the menu OR
4. For “Variable Alarm Mode,” select a standard deviation (sigma) value from the menu

Figure 4.26. Example of Alarm Threshold Configuration

Step 6. Config: Alarm Ringtone

The SAMmobile 150-D provides six-alarm ringtone options as shown in Figure 4.27 below

1. Select ‘Alarm Ringtone’ in Config screen (refer to figure 4.26)
2. Select desired alarm ringtone
3. Apply the ringtone.

Figure 4.27. Select Alarm Ringtone

Step 7. Config: Gain Stabilization Setup

The SAMmobile 150-D provides Gain Stabilization Setup as shown in Figure 4.28 below

Background Limit (CPS)	If the background contains active source(s), then gain stabilization has to be stopped and postponed until background level returns to normal level. 'Background Limit' parameter determines the threshold value of 'active source presence' in the CPS unit.
K40 minimum (Count)	To conduct gain stabilization, a minimum number of K-40 counts are required. 'K40 minimum' parameter determines the lowest threshold value of K40 counts for successful gain stabilization.

Gain Stabilization	
Background Limit (CPS)	3500
K40 minimum (Count)	1500

Figure 4.28. Gain Stabilization Setup

Step 8. Background Measurement

Reliable background information is critical for good isotope identification and alarm performance. The software employs an advanced “NORM Rejection Algorithm” to reduce false alarms due to the fluctuations in the naturally occurring background.

Appropriate background measurement and system calibration of the SAMmobile 150-D will ensure the best possible results. Background Measurement and Manual Calibration are accessed by clicking on the Calibration menu.

1. Click the “Home” menu on the main menu bar
2. Turn off Gauge in the Home Screen
3. Click “Calibration” menu on the main menu bar
4. Select “Background”

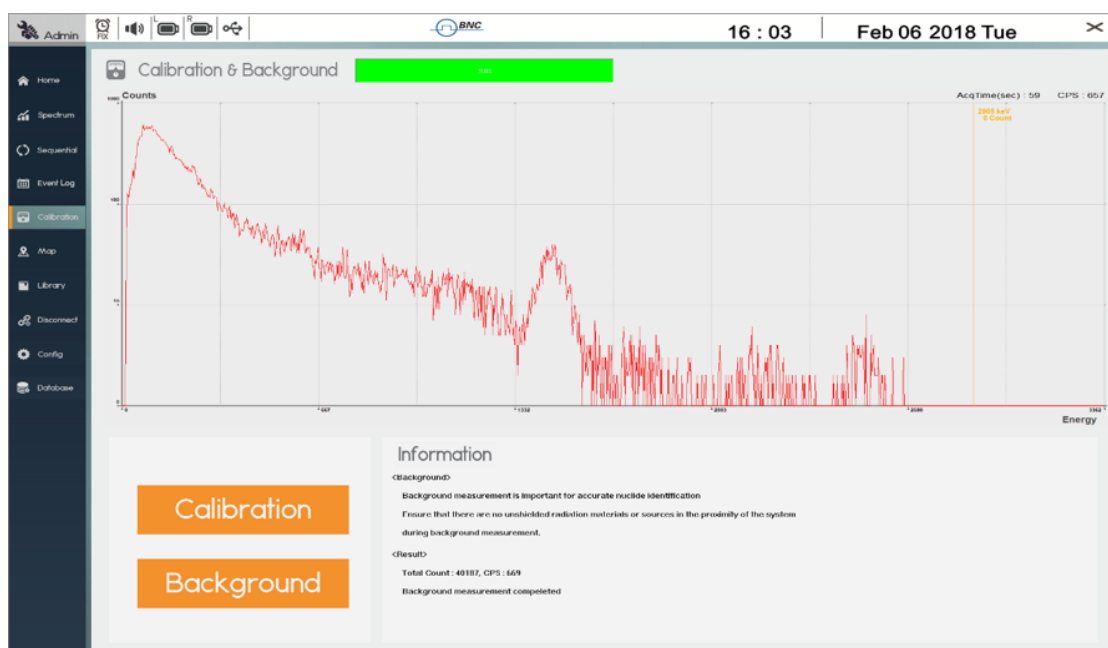


Figure 4.29. Background Measurement

 **Ensure that there are no radiation sources near the SAMmobile 150-D during the background measurement.**



The background spectrum is automatically updated during the measurement. The background measurement time is defined in the “Config” menu.

Step 9. Calibration

1. Click the “Home” menu on the main menu bar
2. Turn off Gauge in the Home screen
3. Place a Cs-137 source on the detector center mark on the top of the SAMmobile 150-D (shown in figure 4.30)

 Note: if you have two Cs-137 sources with similar activities, place them on the detector center as shown in figure 4.30). If only single Cs-137 source is available, place it on between two 150-D units as shown in figure 4.31).

4. Click “Calibration” menu on the main menu bar
5. Select “Calibration”

 Note: The calibration screen can be toggled between left and right 150-D. (shown in figure 4.32)



Figure 4.30. Cs-137 two sources



Figure 4.31. Cs-137 one source

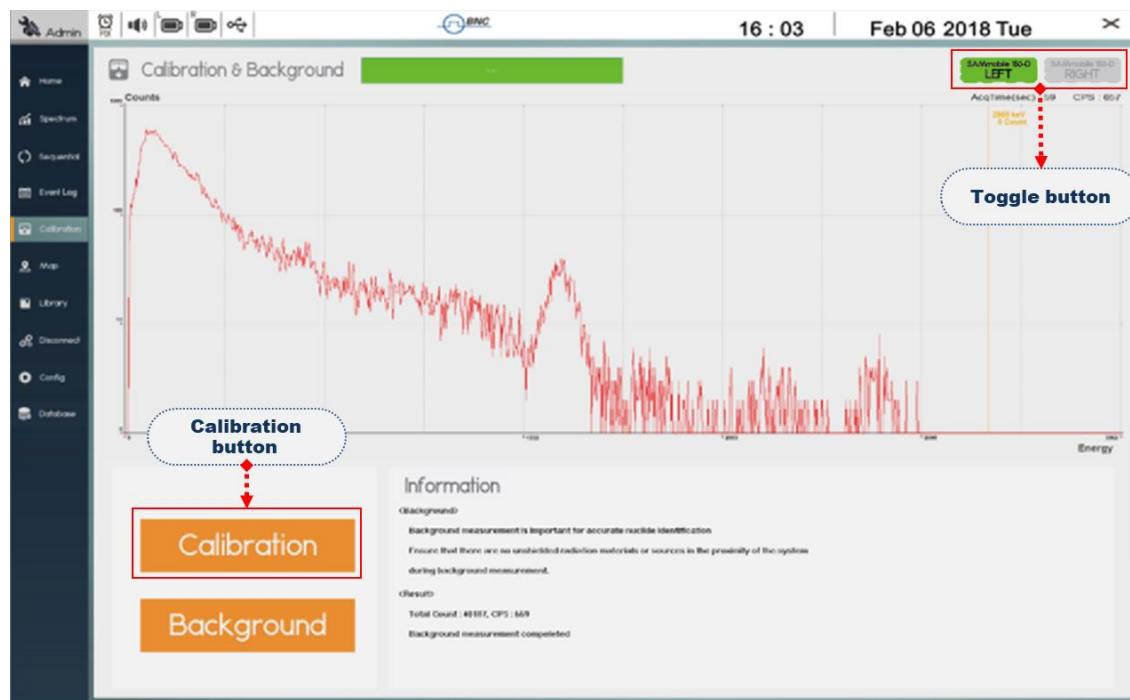


Figure 4.32. Calibration Example

It is recommended that the total integrated counts in the background spectrum should be not less than 100,000. The total Energy Calibration count is defined in the “Config” menu.

Step 10. Radiation Monitoring

Now the SAMMobile 150-D is ready for operation.

1. Turn on “Gauge” in the Home screen
2. Gamma activity is displayed in both dose-rate ($\mu\text{Sv/h}$ or rem/h) and count rate (cps) units.



If “Variable” Alarm has been selected, allow the unit to collect background statistics for about 20 seconds after turning on “Monitor.”



Figure 4.33. Alarm and Radiation Measurement in Progress

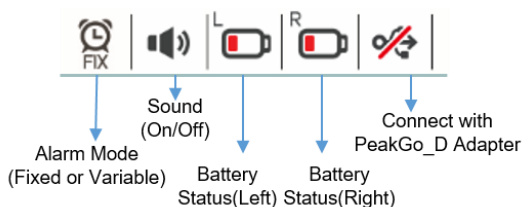


Figure 4.34. Short-Cut Icon Functions

Alarm Mode	Alarm Mode icon shows either FIX or VAR
Bluetooth Search	Search for Bluetooth paired 150-D devices.
Sound	Alarm sound on/off
Battery Status (Left)	Battery status of Left 150-D
Battery Status (Right)	Battery status of Right 150-D
Network connection	Connection status between PeakGo-D and 150-D units.

Gamma activity units are shown as $\mu\text{Sv/h}$ or urem/h (default). The Recent Count Activity graph is logarithmic and auto-scaling.

Several methods have been proposed for the calculation of dose rate using spectral data provided by a NaI(Tl) detector. The SAMmobile 150-D employs a method defined by Kwon *et al.* [1]

Step 11. Alarm & Isotope Identification

If the count rate rises above background (defined by the selected Alarm Mode and Threshold), the system automatically initiates Alarm Mode. When the SAMmobile 150-D alarms, the alarm symbol appears in red (as shown in figure 4.33) and the Alarm Ringtone is activated. Nuclide identification is automatically started. Switching to the “Spectrum” menu will display the nuclide identification data. In this screen, the gathering spectral data, identified isotopes, and statistical confidence levels are displayed as shown in figure 4.35. If the count rate decreases below the Alarm Mode threshold, the alarm symbol and audible warning turn off.



Execute Manual ID to save the event into the Event Log file. Refer to Step 12.

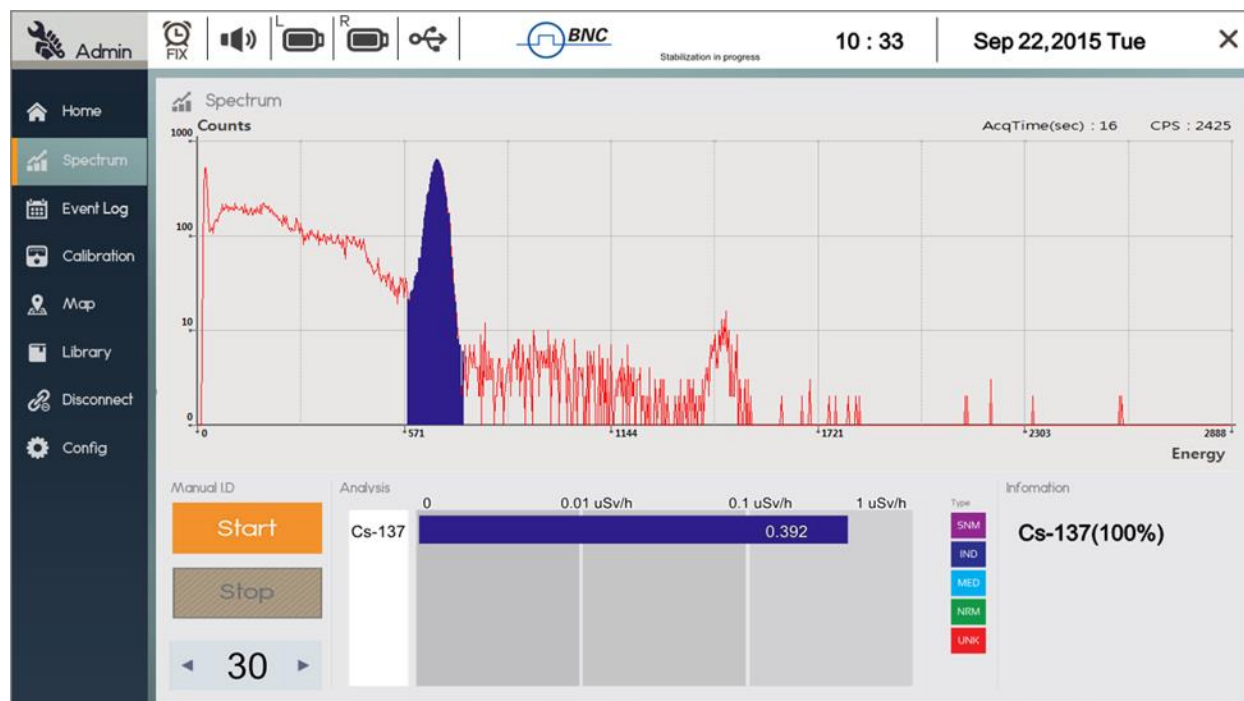


Figure 4.35. Isotope Identification



The required nuclides must be included in the selected radionuclide library.

If multiple isotopes are identified, SAM III PeakGo-D will place the most significant isotope (highest contributing dose rate) as the top-most bar, and the remaining isotopes in descending order to the least significant isotope (lowest contributing dose rate). Each bar will be named and displayed with the corresponding classification color as shown in figure 4.35.

Spectrum	Real-time spectrum display with Date/Time, Acq. Time and gamma count rate information
Counts	Y-axis units (counts in self-adjusting, logarithmic scale)
Energy	X-axis units (energy in keV displayed on a linear scale)
Spectrum	Real-time spectrum display
Analysis	Isotope ID, dose rate, and classification of individual isotope(s) Class of identified isotope SNM: Special Nuclear Material – Magenta

	IND: Industrial Isotopes – Blue MED: Medical Isotopes – Cyan NRM: Naturally Occurring Radioactive Material – Green UNK: Unknown Material – Red (isotope is not in the library)
Information	Statistical confidence of each identification

 When the measured dose rate exceeds the “Health Alarm” threshold, the screen color changes to yellow as shown in figure 4.36, and blinks. The operator(s) should evacuate from the site immediately.



Figure 4.36. Health Alarm Screen

Step 12. Manual Identification

Even when Alarm Mode is in progress, an Identification Event may be performed manually.

1. Touch Start in the Manual ID field and an Event Spectrum will begin immediately.
2. The default capture time for Manual Identification is 60 seconds. You may extend or reduce the capture time by touching the ◀ or ▶ buttons respectively. Time is extended or reduced in increments of 30 seconds.
3. When the capture time has elapsed, event information is automatically saved to the Event Log file. You may touch Stop to cancel the capture at any time.



Figure 4.37. Manual Identification

Step 13. Open Saved Event

The Event Log associated with a Manual ID is automatically generated and saved in the Tablet PC.

Analysis of the spectra for saved events can be performed by the following steps.

1. Turn off Gauge in the “Home” screen
2. Click “Event Log” on the main Menu bar.
3. Select “Event by Date” menu
4. Select the event by number or by time stamp from the “Event List” menu

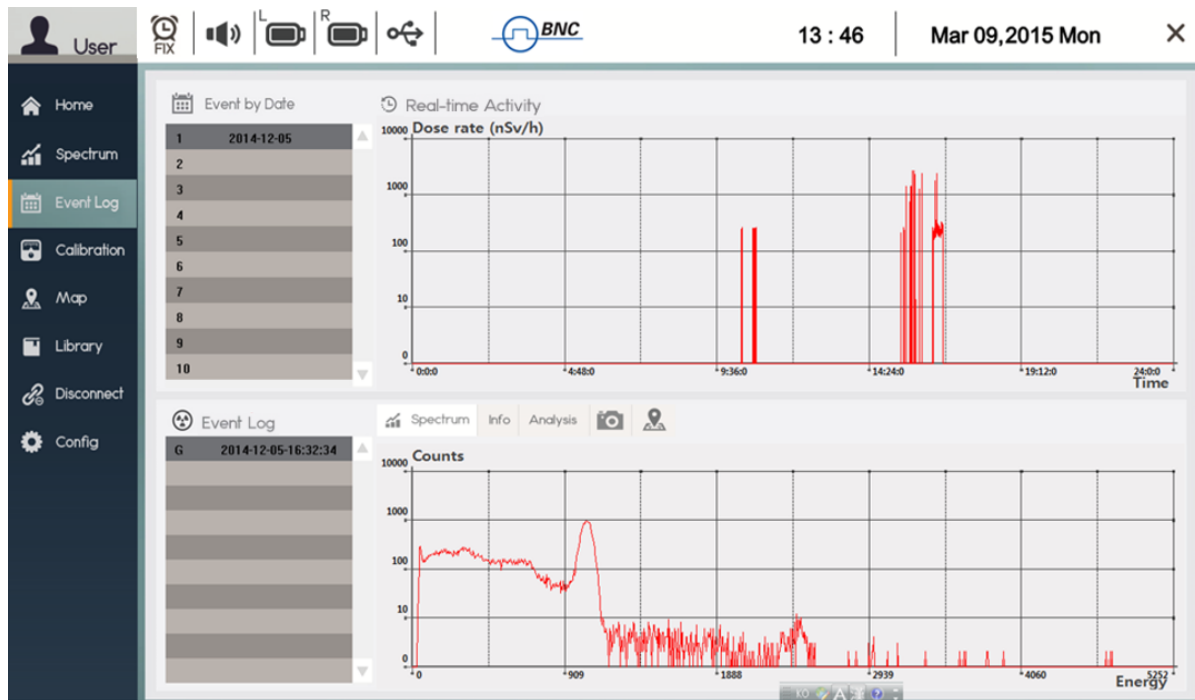


Figure 4.38. Opened Event

Step 14. Event Analysis

An event spectrum and its associated information can be displayed by selecting the event from the Event Log list. The Event Log screen appears as shown in figure 4.39. The Event Log screen contains 5 tabs: Spectrum, Info, Analysis, Photo, and Map.

✓ Spectrum

Spectrum manipulation techniques described in figure 4.19 and 4.20 apply to the event analysis screen.

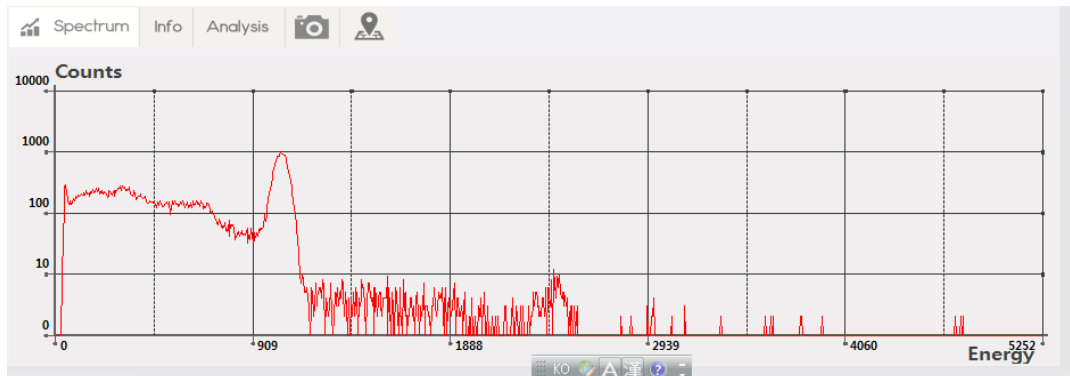


Figure 4.39. Spectrum Tab

✓ Info

The Info tab contains alarm date/time, user, dose rate information, and comments. Adding a comment is described in Step 15: Add Comments and Photo/Video.

Spectrum Info Analysis Photo Map			
Device	HH100	Date	2014-12-05
Time	2014-12-05-16:32:34	Alarm Duration	16 Sec
User	DDD	Location	ABC
Avg. CPS	2979	Total CPS	47665
Avg. Dose rate	200 urem/h	Avg. Neutron	0 CPS
Max. Dose rate	234.8 urem/h	Max. Neutron	0 CPS
COMMENT			
DoseRate			

Figure 4.40. Information Tab

✓ Analysis

In the Analysis tab, the measured dose rate of identified isotopes is graphically displayed in a horizontally oriented bar graph. If multiple isotopes are identified, SAM III PeakGo-D will place the most significant isotope (highest contributing dose rate) as the top-most bar, and the remaining isotopes in descending order to the least significant isotope (lowest contributing dose rate). Each bar will be named and displayed with the corresponding classification color, and with the statistical confidence of each identification as shown in figure 4.41.

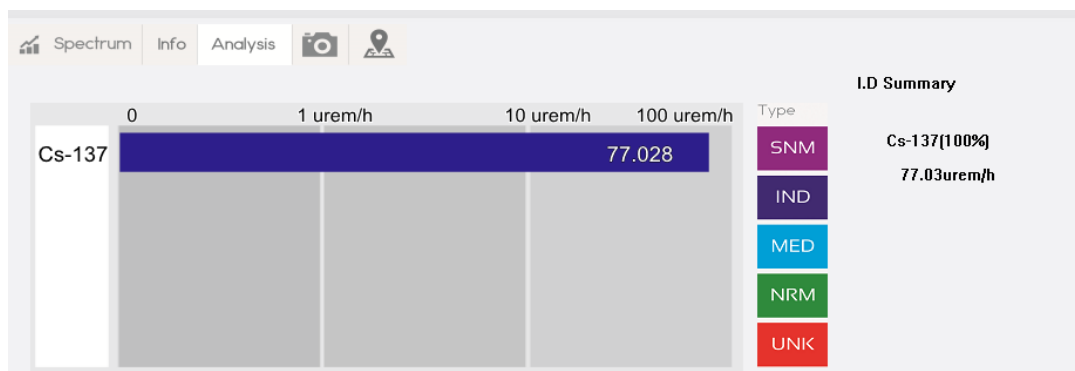


Figure 4.41. Analysis Tab

✓ Photo/Video

In the Photo/Video Tab, information in image form can be saved and attached to each event. Adding photo/video is described in Step 15.



Figure 4.42. Photo/Video Tab

✓ Map

If the PC/Tablet is within range of an internet-enabled network, each event will contain GPS information from which a radiation map can be generated using that information. The Map tab displays an area map of the event location along with dose rate information.

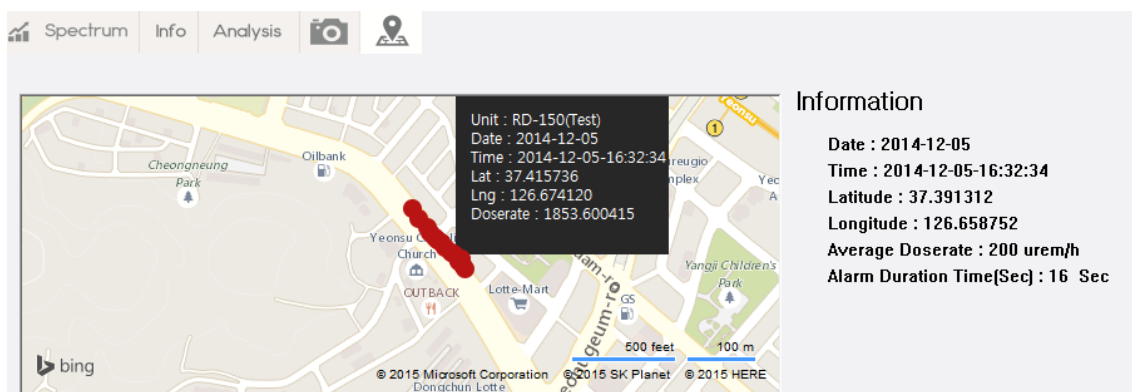


Figure 4.43. Map Tab

Step 15: Add Comments and Photo/Video

Comments and/or images may be attached to the opened event.

“Attach Comment” procedure is shown in figure 4.44.

1. Click the “Info” tab.
2. Touch the comment area. The keypad and typing window screen will appear.
3. Type comments into the field and touch the “Apply” button.



Figure 4.44. Attach Comment

“Attach Photo/Video” procedure is shown in figure 4.45.

1. Touch the  tab.
2. Touch “NEW MEDIA” bar. The Photo / Video application will open.
3. Capture photo or video images. They are automatically saved when image capture is completed.

 The operator can save as many photo/video images as needed. The total number of photo/video images saved is displayed on the screen.



Figure 4.45. Attach Photo

AUTO-CALIBRATION AND STABILIZATION

The Auto Calibration and the Stabilization features adjust the calibration parameters of the system to correct for any drift caused by environmental changes (generally ambient temperature).

Auto Calibration is executed automatically when the Bluetooth pairing of the SAMmobile 150-D and the Tablet/PC is successful. If the operator chooses to cancel the Auto Calibration, it can be stopped by touching the 'Cancel' button. Stabilization is automatically executed every 5 minutes unless the unit is in Alarm Mode.

When Auto Calibration and Stabilization are executed, the system automatically adjusts its gain using the 1.4 MeV energy peak generated by the K-40 radionuclide of naturally occurring potassium found nearby in the local environment. A calibration source is not needed. The stopping criteria for this automatic feature are the integrated counts in the K-40 peak. The total Energy Calibration count is defined in the 'Config' menu. Typically 100,000 counts are recommended to ensure that the K-40 stabilization peak is well defined.



Supplemental Potassium Chloride (KCL) salt can be used to assist the Stabilization if necessary.



If a drift of more than $\pm 6\%$ between the stabilized peak and the original reference peak occurs, then Manual Calibration using a Cs-137 source may be required.



Typically, when Bluetooth pairing between the SAMmobile 150-D and the Tablet/PC is complete, PeakGo-D will immediately perform Auto Calibration. It is advisable to wait for Auto Calibration to complete (100%) before proceeding further. If the system has been calibrated manually recently, then the operator may cancel before completion.

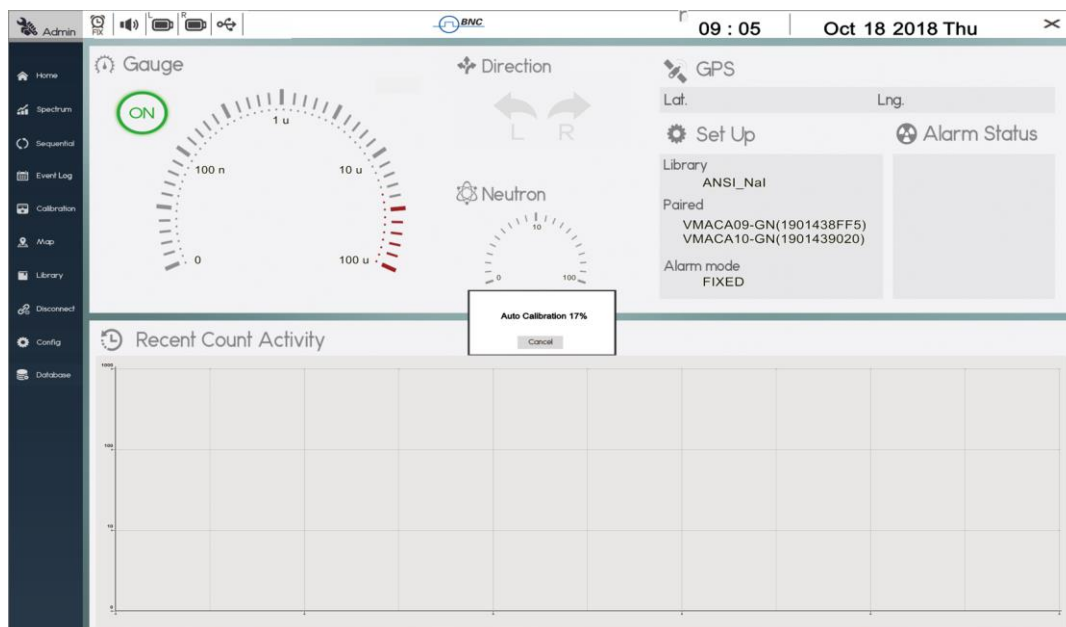


Figure 4.46. Auto Calibration

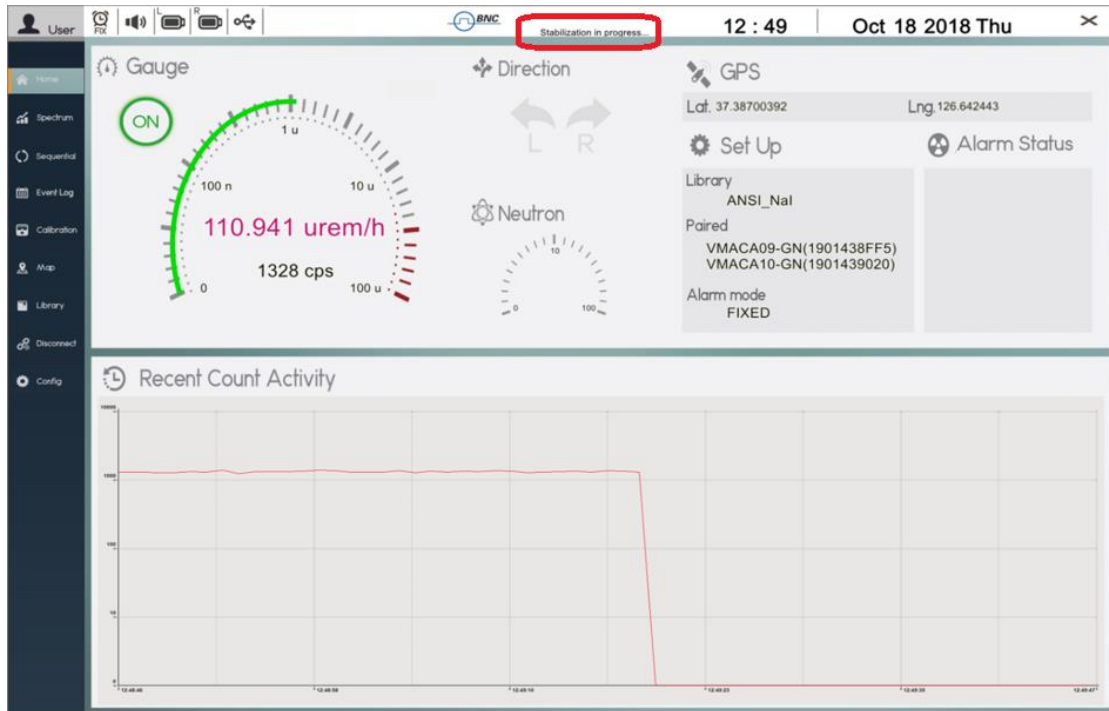


Figure 4.47. Stabilization

CHAPTER 5. REMINDER AND USEFUL TIPS

- ✓ Do not open the cover. The servicing of the unit should only be handled by trained personnel authorized by Berkeley Nucleonics Corp.
- ✓ Turn off unnecessary PC applications to optimize the performance of the Tablet PC, and to optimize communication and operation of the SAMmobile 150-D.
- ✓ Be familiar with the Tablet PC and Windows operation. Please refer to the Samsung Tablet PC user manual.

References

[1] Calculation of Neutron and Gamma-Ray Flux-to-Dose-Rate Conversion Factors, S Kwon, K Kim, C. Ha, S. Moon, and C Yook, Journal of the Korean Nuclear Society Vol. 12, N 3, September 1980.

Appendix: SAMmobile 150-D Specification

INPUT/OUTPUT	
USB	Micro USB 2.0
Bluetooth	Class 2.0, max 100m range (class2)
GPS	Built-in GPS in a Tablet PC or Smartphone
PHYSICAL	
Dimensions (W x D x H)	84 x 62 x 41 (cm): VM100-D-2G2 or VM100-D-4G2 model 108 x 34 x 30 (cm): VM100-D-2G1 or VM100-D-4G1 model
Weight	25 kg (49 lb) VM100-D-2G1 model
ENVIRONMENTAL	
Operating Temperature	5 to 122 F (-15 to 50 C)
Relative Humidity	10 to 80%, noncondensing
Protection Rating	IP 65 (according to IEC 60529)
PERFORMANCE	
Energy Resolution (Gamma)	NaI(Tl) 2x4x16 inch: > 8% @662 keV
Energy Range (Gamma)	20 keV – 3 MeV
Throughput	> 150 kcps
MCA channel	10bit 1024 channel
Dose rate range (NaI)	0 – 10 mR/h
Dose rate range (GM)	10 mR/h – 10 R/h (optional)
Stabilization	Automatic real-time stabilization using K-40
Nuclide Identification	According to ANSI N42.34, isotope/category/confidence report
Battery	> 8 hours, Lithium Ion
DETECTORS	
Gamma NaI(Tl)	2x4x16 inch – Standard, 4x4x16 inch – optional, up to 2 detectors/unit
Gamma (High Dose Rate)	Geiger-Muller detector (Optional)
Neutron (optional)	Solid-state Neutron detector: 4 cm ² active area, 30% thermal neutron eff. Gamma rejection: 1: 10 ⁷ . He-3: 27 cps/nv sensitivity.
DISPLAY	
Smartphone	Samsung Galaxy Player or equivalent
Tablet PC	Any Windows compatible tablet PC
SOFTWARE	
Application SW	Android-based application SW for Smartphone (SAM III PeakAbout-D) Windows-based application SW for Tablet PC (SAM II PeakGo-D)
Reach-back Feature	ANSI N42.42 event data via a Tablet PC or Smartphone



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